TEC Control: Oral Exam Study Guide

Modules 2–6 | sensor, actuator, P/PI feedback, and thermal models

A connected explanation using the Naidu/Cohen bench data. Equations are written one step per line; each step is followed by what the math means physically.

The story in one minute

Module 2 turns thermistor voltage into temperature and checks the H-bridge signal path. Module 4 measures how signed PWM changes the plate temperature. Module 5 uses temperature error to choose PWM. Module 6 writes the heat balance as a differential equation, then adds integral action and explains why extra thermal states can produce behavior a one-lump P model cannot.

A useful oral answer pattern: state the equation, define the symbols and units, show the algebra, then connect the result to a specific measurement and state one limitation.

1. Module 2 — temperature measurement

The circuit is a voltage divider: a 100 kΩ fixed resistor is the upper leg and the NTC thermistor is the lower leg. The Arduino reads the midpoint voltage at A0.

$$V_{A0} = V_{ref} \cdot R_{th} / (R_{th} + R_{fixed})$$

As the NTC warms, its resistance falls. Since it is the lower divider leg, A0 voltage falls; the conversion then reports a higher temperature. The finger test confirmed this direction.

The conversion chain

1. Average N ADC counts: $\bar{n} = (n_1 + n_2 + \ldots + n_N) / N$
2. Convert the average count: $V = V_{ref} \cdot \bar{n} / 1023$
3. Invert the divider: $R_{th} = R_{fixed} \cdot V / (V_{ref} - V)$
4. Beta model: $1/T_K = 1/T_0 + (1/B) \ln(R_{th}/R_0)$
5. Convert to Celsius: $T_C = T_K - 273.15$

Worked example from the Module 2 log: 450.3 counts, 5.00 V reference, and 100.0 kΩ fixed resistor.

$$V = 5.00 \cdot 450.3 / 1023 = 2.201 \text{ V}$$
$$R_{th} = 100.0 \text{ k}\Omega \cdot 2.201 / (5.00 - 2.201) = 78.64 \text{ k}\Omega$$
$$T = [1/298.15 + (1/4540) \ln(78.64/100.00)]^{-1}$$
$$T = 302.95 \text{ K} - 273.15 = 29.8 \text{ °C}$$

Why average before converting? The Beta conversion is nonlinear. Therefore, average(convert(each reading)) is generally not equal to convert(average count). The assignment's method averages ADC counts first, then converts once. Averaging reduces random variation; it does not fix a miswired divider, a wrong B value, or reference-voltage error.

2. Module 2 — PWM and the H-bridge

PWM rapidly switches an output on and off. The duty fraction is the fraction of each period spent on:

$$D = t_{on} / t_{period}$$

For a 0–255 command: D is approximately PWM / 255

PWM 0 means no commanded drive; 255 is continuously on. The bridge selects direction by energizing one input while holding the other low. The bench calibration found pin 10 heats and pin 9 cools, so PIN9_IS_HEAT is false. The Module 2 load was a small motor; the TEC was disconnected for that test.

Scope safety: M+ and M− are driven H-bridge outputs, not ground. The scope ground clip goes to Arduino GND; clipping it to a driven output can short the bridge.

3. Module 4 — open-loop calibration

Open loop means the operator chooses PWM and direction; temperature does not automatically choose the next command. The measured susceptibility is temperature change per signed PWM count.

T_{ss} is approximately T_{amb} + chi*u
chi = change T / change u (°C per PWM count)
u > 0 means HEAT; u < 0 means COOL

Branch	Measured slope magnitude	Interpretation
Heating	$\chi_h = 0.50127$ °C/count	signed slope is positive
Cooling	$\lvert\chi_c\rvert = 0.18091$ °C/count	temperature falls as negative u grows

The slopes differ because Peltier transport reverses with current, but resistive Joule heating does not. Passive heat transfer always pushes the plate toward room temperature.

PWM averages and the Peltier/Joule distinction

During each period tau: I(t) = I for D*tau; otherwise I(t) = 0
Average I = [I*(D*tau) + 0*((1-D)*tau)] / tau = D*I
Average I squared = [I^2*(D*tau) + 0*((1-D)*tau)] / tau = D*I^2
But (average I)^2 = D^2*I^2, which is not average I squared = D*I^2

Peltier transport is approximately proportional to average I; Joule heating is proportional to average I squared. Under this on/off PWM model both are proportional to duty D, not D^2. A slope ratio from Module 4 is:

$r = m_h / \lvert m_c\rvert = 2.771$
$Q_J/Q_P = (r - 1)/(r + 1)$
$Q_J/Q_P = (2.771 - 1)/(2.771 + 1) = 0.470$

Treat r as approximately 2.77 ± 0.05, not exact to four figures. Several Module 4 steps had not reached steady state, and the +50 point has no saved run. The direction calibration is stronger than the precise slope ratio.

4. Module 5 — P control, Kp, and droop

The controller compares the target with the measured plate temperature, then multiplies error by proportional gain.

$e = T_{set} - T$
$u = K_p e$
K_p units = PWM counts/°C; u units = PWM counts
sign(u) selects HEAT or COOL; |u| is rounded and clamped to 0–255

Kp is not a target temperature and does not directly specify °C/s. It specifies how many PWM counts are requested for each °C of error. Higher Kp usually drives harder and reduces droop, but can overshoot or ring if the real system has lag.

Step-by-step steady-state droop derivation

Measured open-loop relation: $T = T_{amb} + \chi u$
Substitute P control: $u = K_p(T_{set} - T)$
$T = T_{amb} + \chi K_p(T_{set} - T)$
Define dimensionless loop gain: $L = \chi K_p$
$T = T_{amb} + L\,T_{set} - LT$
Move $-LT$ to the left: $(1 + L)T = T_{amb} + LT_{set}$
Solve: $T_{ss} = (T_{amb} + LT_{set})/(1 + L)$
Droop: $T_{set} - T_{ss} = (T_{set} - T_{amb})/(1 + L)$

Physical meaning: at equilibrium, TEC heat input balances passive heat loss. If L is small, much of the initial error remains. If L is large, less error remains. P alone needs nonzero error to command nonzero steady PWM; zero error would mean zero P output.

Worked prediction from the established Sept 30 Module 5 run: Kp = 2, Tset = 30.0°C, Tamb = 21.54°C, $\chi h = 0.50127$°C/count.

$e_0 = 30.00 - 21.54 = 8.46$ °C
$L = 0.50127 \times 2 = 1.00254$
Predicted droop $= 8.46/(1 + 1.00254) = 4.2246$ °C
Predicted Tss $= 30.00 - 4.2246 = 25.7754$ °C, about 25.78 °C
Measured final-minute mean = 25.75 °C; difference is about −0.03 °C

Your Oct 13 repeat is a different dataset: only Kp = 16 and 32 met the 0.16°C/min drift check. Do not call lower-gain final-minute means steady-state values.

5. Module 6 — dynamic model and time constant

The one-lump model is an energy balance: stored-energy rate equals TEC heat input minus passive heat loss.

$C\,dT/dt = P_u u - H(T - T_{amb})$
C: thermal capacitance (J/°C); H: heat-loss conductance (W/°C)
P_u: effective TEC power per PWM count (W/count)

Divide by H and define measured parameters $\chi = P_u/H$ and $\tau = C/H$. This removes the separately unknown C, H, and P_u:

$dT/dt = [T_{amb} + \chi u - T]/\tau$
Units: χ = °C/count; τ = seconds

For constant u, define the equilibrium T_final = T_amb + chi*u and departure theta = T − T_final. Substitution reduces the model to:

$d(\text{theta})/dt = -\text{theta}/\text{tau}$
Separate variables: $d(\text{theta})/\text{theta} = -dt/\text{tau}$
Integrate: $\ln|\text{theta}| = -t/\text{tau} + \text{constant}$
Apply theta(0) = theta_0: theta(t) = theta_0*$e^{-t/\text{tau}}$
Therefore: $T(t) = T_{final} + [T(0) - T_{final}]e^{-t/\text{tau}}$

At t = tau, the remaining departure is exp(−1), about 0.368 of its initial value, so about 63.2% of the total step has occurred. Your fitted open-loop time constant is $\tau = 62.9 \pm 5.0$ s.

Closed-loop P dynamics

Substitute $u = K_p(T_{set} - T)$ into the energy balance
For theta $= T - T_{ss}$: d(theta)/dt $= -(1 + L)$*theta/tau
Closed-loop time constant: tau_cl $=$ tau/$(1 + L)$

There is one state and one real negative eigenvalue: lambda $= -(1+L)/\tau$. The exponential keeps its sign while shrinking, so an ideal one-lump P response cannot cross its final value or oscillate. Measured overshoot means the real apparatus has dynamics this model leaves out.

6. PI control, matrices, and windup

PI adds a memory of past error. Let q be accumulated error:

q(t) $=$ accumulated error (area under e versus time); dq/dt $=$ e
$u = K_p e + K_i q$
Discrete update: $q_{n+1} = q_n + e_n$*delta_t
Then clamp the requested u to the allowed signed PWM range

At steady state, an unsaturated PI loop can have $e = 0$ while the integral term supplies the nonzero PWM needed to balance heat loss. That is why PI can remove P droop.

PI state matrix around equilibrium

Let theta $= T - T_{ss}$; z $=$ integral-state deviation
d(theta)/dt $= -((1+L)/\tau)$*theta + (chi*K_i/τ)*z
dz/dt $= -$theta
State matrix A $= [[-(1+L)/\tau, \text{chi}*K_i/\tau], [-1, 0]]$
Characteristic equation: lambda^2 + $((1+L)/\tau)$*lambda + chi*$K_i/\tau = 0$
zeta $= (1 + L) / [2*\text{sqrt}(\tau*\text{chi}*K_i)]$

The eigenvalues are roots of the quadratic. Two negative real roots give an overdamped response; a complex-conjugate pair with negative real part gives a decaying oscillation. Increasing Ki lowers zeta and can increase speed and overshoot.

On Oct 13, at fixed PI Kp $= 12$, Ki $= 0.05$ was slow and still creeping, Ki $= 0.2$ reached near-setpoint with little overshoot, and Ki $= 1.0$ removed steady error but overshot 0.71°C on heating and 1.30°C on cooling. This is the speed-versus-overshoot tradeoff.

Windup happens when PWM is clamped but the integral continues accumulating error. When the plate nears the target, stored integral can keep commanding the old direction and cause overshoot. Conditional integration stops accumulating when saturation and error would push farther into the limit.

The 50-count windup demonstration was simulated, not the actual bench power limit. Plain PI overshot by 1.46°C; anti-windup reduced that to 0.26°C.

7. What the data can and cannot say

Evidence	What it supports	Limit / honest qualification
Module 2 thermistor	Divider response and Beta conversion produce plausible temperature	Beta curve has tolerance and is less accurate outside its fitted range
Module 4 PWM slopes	Heating and cooling have different susceptibilities	Several steps were not settled; +50 point has no saved run
Module 5 P	Higher gain reduced droop; Oct 13 Kp 16 and 32 passed drift check	Oct 13 lower gains were still drifting
Module 6 P/PI	PI removes droop; larger Ki traded speed for overshoot	Physical P runs used Kp 2/16/64; PI runs used Kp 12
One-lump model	Predicts droop and basic first-order time scale	Cannot produce overshoot; residuals indicate missing dynamics

Record discrepancy to verify before the oral: the A3 draft describes a matched $K_p = 2$ P/PI comparison, but the saved Oct 13 bench log lists P runs at $K_p = 2, 16, 64$ and PI runs at $K_p = 12$. State the actual logged settings unless your team confirms another matched run exists.

Short oral answers

Why does P control need droop? P output is Kp times error. Zero error gives zero P output, but heat loss still exists, so a nonzero error is needed to request sustaining heat.

What does increasing C do? It increases $\tau = C/H$ and slows the transient; it does not change the steady equilibrium.

What does increasing H do? It increases passive heat loss, decreases $\chi = P_u/H$, and changes $\tau = C/H$.

Why does the one-lump model not ring? It has one state and one real negative eigenvalue. A single decaying exponential cannot cross its equilibrium.

Why can PI overshoot? The integral stores past error. A large Ki changes the two-state eigenvalues toward an underdamped pair; saturation can also cause windup.

What is the strongest evidence for a second time scale? Structured residuals in open-loop traces and physical P overshoot. Sampling alone cannot explain open-loop residuals.

Is Kp a temperature or a speed? Neither. Its units are PWM counts per °C; it converts current error into actuator command. The plant and thermal mass determine the temperature rate.

Primary sources: Module 2 instrument note; Module 4 open-loop note and calibration data; Module 5 P-control note and Oct 13 repeat snapshot; Module 6 modeling note, bench log, and Goldilocks metrics. This guide is a study aid; for discrepancies, use the raw run log and confirm with your instructor.